

Theo Usher

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EDUCATION

Columbia University, *Bachelor of Science, Mechanical Engineering* New York, NY; Sep 2020 – May 2024
GPA 4.07/4.0, TBP Engineering Honor Society, Mechanical Engineering Certificate of Merit, Magna Cum Laude
Electives: Managing Tech Innovation, Robotics Studio, Human-Centered Design, Public Speaking

WORK EXPERIENCE

Boeing, *Mechanical Product Design Engineer* Los Angeles, CA; Aug 2024 – Present

- Lead mechanical design, integration, and serial production support of spacecraft hardware assemblies
- Develop unconventional integration solutions to meet electrical, thermal, and spatial constraints
- Work with diverse manufacturing methods to create electronics packaging for aircraft and spacecraft
- Design fixtures, jigs, and software tools to improve production and evaluation efficiency
- Acted as production support lead, coordinating engineers and dispositioning millions in flight hardware
- Recognized by team lead for maintaining production schedule and quality despite elevated demand

Airobotics Drones, *Design Engineering Intern* Tel Aviv, Israel; May 2023 – Jul 2023

- Designed mechanical systems for the Iron Drone, an autonomous counter-drone
- Analyzed, tested, and optimized the drone's net launcher for range and reliable interception
- Redesigned the drone ground station for improved functionality, reliability, and environmental protection
- Optimized the drone ground station, cutting assembly time from 2 weeks to 2 days
- Designed and manufactured jigs to reduce friction points in drone manufacturing and system integration
- Engineering contributions used in Iron Drone systems sold globally for \$25M+ and exhibited by the U.S. DOT

Terabase Energy, *Mechanical Engineering Intern* Davis, CA; May 2022 – Aug 2022

- Designed and field-tested robotic vehicles to transport and install solar panels in utility-scale projects
- Designed a solar panel installation mechanism that more than doubled installation range
- Built a panel stabilization system that replaced high-cost linear actuators with low-cost, durable mechanisms
- Engineering contributions used in automated solar installation factories that have installed 40+ MW of panels

LEADERSHIP & ENGINEERING CLUBS

Columbia Space Initiative, *Rockets Mission Lead* Sep 2020 – Jun 2024

- Led a 50-person team to research, design, and build a hybrid rocket to reach 30,000 ft with a scientific payload
- Developed a dual-track design strategy of bold, innovative systems supported by low-risk backups
- Led cross-subsystem integration for both design and physical assembly
- Led manufacturing across a range of processes including CNC machining, 3D printing, and sheet metal forming
- Designed, manufactured, and tested parachute deployment mechanisms for rocket recovery
- Created and ran test procedures, including for rocket static fires and recovery testing
- Achieved Rockets' first successful flight and recovery, improved engine performance by 10x
- Cultivated a team that, a year later, became the first US student-led group to launch a liquid-oxygen hybrid rocket

ENGINEERING DESIGN PROJECTS

Automated Robotic Linkage – Machine Design Sep 2023 – Dec 2023

- Designed, built, and programmed a complex linkage mechanism to rapidly press arcade buttons
- Created a unique, cable-driven mechanism unlike any that our professor had seen, greatly reducing inertia

Bipedal Walking Robot Jan 2022 – May 2022

- Designed, manufactured, and programmed a bipedal robot using SolidWorks, 3D printing, and a Raspberry Pi
- Achieved the first and fastest walking robot in a class of significantly more experienced engineers

SKILLS

Technical Skills: CAD (SolidWorks, Creo, Fusion), 3D Printing, CNC Machining, Sheet Metal Forming, FEA, GD&T, CAM, DFM, DFA, PDM, MATLAB, Python, C++, Root Cause Analysis, Production Support

Non-Technical Skills: Project Management, Cross-Functional Collaboration, Public Speaking